

SSC CGL (COMBINED GRADUATE LEVEL - INSPECTOR, ASO, TAX ASST) — SUBJECT COMPLETE QUESTION BANK

PRACTICE MATERIAL • VERIFIED QUESTION BANK

Total Questions: 176 • Mode: Single Subject Comprehensive Practice

- Q.1.** '©@'%@ 'YG 'éB")M'ò .y 20% 'Y@ "YC'iM'y? "TMK'%G '©0 •ù 'yC"TM#" "2 '©(" '©\$ 'éG•" "ù\$'%G '©M" \$"ù6'B " 'é@ 'Y0"
A) 16.67% (16 !T%) B) 20%
C) 25% D) 15%
- Q.2.** •ù 'iA'Y>%&"é0 'Y?'%@ "Y8"ù\$" * " # R " R " &"2 "Ù0'é?'R ") (Successive Discounts) 'iG'I> "TMH-B 'Y2 "%0.'IA")M'ò , 'ùM"
A) 28% B) 30%
C) 25% D) 27%
- Q.3.** •ù 'ÉG•%0."é("YM'ù>'©>" @ •Y*'%@ "Y8"ù\$" • " "2 "Ù0'ò .")2"Ù/ '©0 'ÉG'©("r "â &"é5"â " \$"â 9", * " 'IA 1 'Y?'yM" > 'YG "%0M"
A) 11.11% (11 !Q%) B) 10%
C) 12.5% D) 9%
- Q.4.** 'YK•, "%00"é6"ò 8"é"é0'2 , "Ù/"é 'Y@ 'i0 "%0G 8 "Y0"Ù7"1 'éG•" &"1 " (" 9"2 "é\$" 9"%0d "Y9" "%00"é6"ò "ù\$'%G "Y0"Ù7"1 'éG•"
A) 24 "Y0"Ù7 / 24 Years B) 16 "Y0"Ù7 / 16 Years
C) 32 "Y0"Ù7 / 32 Years D) 20 "Y0"Ù7 / 20 Years
- Q.5.** 110,000 'Y@ " >"i? '©0 10% "Y>" M"y?'R & " 8"r " 5" M"r "r 'YM" 5"9&"ù"ò , "Ù/"é 'i0 "%0>'y>" # 'ÉM'ù>Â "â •)\$" „F-f-erence
A) 1100 B) 1150
C) 1200 D) 150
- Q.6.** 'ù&"ò B = 2 : 3 'l%"â " C = 4 : 5 "TMK, 'IK A : B : C 'Y> •Y(" *é\$ 'YM'ù> "TMK'y>?
A) 8 : 12 : 15 B) 6 : 9 : 15
C) 8 : 10 : 15 D) 4 : 6 : 10
- Q.7.** A 'Y?'%@ 'Y>" M'ò "2 &"ù('éG•" \$'Y> B •TM8" "é0"Ù/ 'YK 15 'i?', .y '©B" > 'Y0 "%0 'l> "TMH-B &"1("1 'é?) " " , "é0"Ù/ 'YK 'Y"
A) 6 'i?', ò b F ys B) 8 'i?', ò , F ys
C) 5 'i?', ò R F ys D) 7 'i?', ò r F ys
- Q.8.** '©>•Y* A •ù 'ù 'Y@ 'YK 6 '%0 'ùG 'éG•" -" 8'Y\$"â 9", " *é '¢ " "%0G 8 '%0 'ùG 'éG•" "é2" " 8'Y\$"â 9"%0d 'iK'%0K•" *é '¢ 'R 8"é"
A) 24 '%0 'ùG / 24 Hours B) 14 '%0 'ùG / 14 Hours
C) 18 '%0 'ùG / 18 Hours D) 12 '%0 'ùG / 12 Hours
- Q.9.** 180 'é@'ù0 ") 'É@ •ù " G") "é!"É@ 54 'Y?'é@/'%0 'ù> 'Y@ '©>" " 8"r 'R •)-"r "2 "ù\$'%G "%0G'Y ' .y '©>" " G'y@?
A) 12 "%0G'Y ' ò " 6V6öæG0 B) 10 "%0G'Y ' ò 6V6öæG0
C) 15 "%0G'Y ' ò R 6V6öæG0 D) 18 "%0G'Y ' ò , 6V6öæG0
- Q.10.** "i>•)\$ 'É2 'éG•" 'R ("é5 'Y@ '©>" " "ù." ù •) "â 9", \$'Y> 'y>" > 'Y@ '©>" " "ù." ù •) "â 9"%0d 'y>" > 'YG •Y(" ")2 (Downstream)
A) 3 '%0 'ùG / 3 Hours B) 4 '%0 'ùG / 4 Hours
C) 3.6 '%0 'ùG / 3.6 Hours D) 4.5 '%0 'ùG / 4.5 Hours
- Q.11.** '©M" %"â S * "Ù0"é "9\$"ù "%0 'iM'ù>'9 'Y> 'l8'B "Ù/"â 9"1 "ãð
A) 25.5 B) 25
C) 26 D) 50
- Q.12.** '©?'l> 'i0 '©A'IM" " 5" M'l."é(•i/" "â '%0A'©>'B R 2 "TMH-B B 5" M"r , "é& •TM('Y@ •i/" "â /"1 43 "Y0"Ù7 "TMK'y>-B *ù\$"â " "
A) 25 "Y0"Ù7 / 25 Years B) 30 "Y0"Ù7 / 30 Years
C) 35 "Y0"Ù7 / 35 Years D) 20 "Y0"Ù7 / 20 Years
- Q.13.** 'ù&"ò 8•) "Ù/"â C f 3cs2 ("Â f" 8"r *)0"Ù#I "Y?'ù>'ÉM'ò 9"2Â \$"2 v2r "r 8"Ù%"é('©0 'YL', 8"â •) •i 'y>?
A) 7 B) 2
C) 5 D) 6
- Q.14.** 'yA'9('12 (7 y u - 3 u x) 'éG•" 'Y>•, "â •) (Unit Digit) 'YM'ù> "TMK'y>?
A) 4 B) 6
C) 3 D) 7

- Q.15.** 'iK "% 'iM'ù>'9 'Y> •Y(" *é\$ 3 : 4 "™H 'l%"â '% "â .-i8-b ,,,4b' B 9"%d •™('Y> ")f"%f (LCM) 'YM'ù> "™K'y>?
A) 48 B) 36
C) 24 D) 60
- Q.16.** 140 '©M" \$"ò "ù "Ù0"â 5"é2"r "é5"" "2 "c *™Ù0'I? 'Y?'yM" > "Y>")G '©>"Y2 'YG "%>'R "ù8 •Y(" *é\$ 'éG•" .ù2"é/"â "é 'Y? 'é
A) 3 : 1 B) 1 : 3
C) 2 : 1 D) 3 : 2
- Q.17.** •ù "YC'IM'B " * " ?'y? 44 "%G'é@ "™H-B "% "â "Ù7"y\$"Ù0'12 (Area) 'YM'ù> "™K'y>?
A) 154 "%G'é@² / 154 cm² B) 308 "%G'é@² / 308 cm²
C) 77 "%G'é@² / 77 cm² D) 616 "%G'é@² / 616 cm²
- Q.18.** •ù ") 'Â 5"9\$"Ù\$ / "i 'YA 'YG •i"é0 'Y@ 'IM" ?'ÉM'ù> 7 "%G'é@ 'l%"â •) "é 24 "%G'é@ "™H-B 6•) " "â 5'YM" *97"Ù " / 'YM
A) 550 "%G'é@² / 550 cm² B) 500 "%G'é@² / 500 cm²
C) 616 "%G'é@² / 616 cm² D) 450 "%G'é@² / 450 cm²
- Q.19.** A 'lO B '%G 'YM" .i 120,000 'lO 130,000 ") "é " 'R 5"Ù"é*é0 "iA" B 'Y?'ù>-B 5" M"r "r •)\$ 'éG•" " 2 115,000 'Y> ")>'Ò 9" -
A) 19,000 B) 16,000
C) 17,500 D) 110,000
- Q.20.** 'ù&"ò , 2 ÷, Ò B 9"2Â \$"2 <" 2 ÷<" "â . "é('YM'ù> "™K'y>?
A) 14 B) 16
C) 18 D) 12
- Q.21.** 'ù&"ò F â , = 4/3 "™K, 'lK sin ,; , "â . "é('YM'ù> "™K'y>?
A) 4/5 B) 3/5
C) 5/4 D) 3/4
- Q.22.** "YM'ù 'É 24 ÷ 4 × (3 + 1) - 6 + 2 'Y> "%0")\$'â . "é('YM'ù> "™K'y>?
A) 20 B) 18
C) 24 D) 16
- Q.23.** 'iK "YM'ù "Ù\$"ò 'R 9" 8"Ù%"é("%G 'YM" .i 4 'Y?'é@/'% "ù> 'lO 5 'Y?'é@/'% "ù> 'Y@ 'y\$"ò 8"r 5"ù*"" @'B &"ù6"é •" .y '©2'lO
A) 27 'Y?'é@ / 27 km B) 15 'Y?'é@ / 15 km
C) 3 'Y?'é@ / 3 km D) 30 'Y?'é@ / 30 km
- Q.24.** •ù '©A"%>"R .y 'iK •™."Ù." &"Y>" %"yd 'É@'l("r 5"é2"r 'ém'é@'i5"é0 'YK 'YA"" 5"% 'é\$"1 'Y> 60% 'é\$ 'é?")> 'lO "Y9 1200 '
A) 6,000 B) 5,000
C) 7,200 D) 4,800
- Q.25.** A 'lO B 'Y?'%>@ 'Y>" M'ò "2 "Ù0'é6•2 b &"ù('lO 8 'i?', .y 'Y0 "% 'lG "™H•)d '©B" G 'Y>" M'ò " . 'É&")0" " C 9"%d A 'Y> "™
A) 1800 B) 1600
C) 1700 D) 1750
- Q.26.** 'ù&"ò 'R 5"9\$"Ù\$ 'lO •ù "Y0"Ù 'Y> '©0"ù. "é* "% . "é("™K, 'lK •™('YG 'YM"yG'IM" +")K•" "â '%A'©>'B "Ù/"â 9"1 "ãð
A) 14 : 11 ("YC'IM'B "â , '<"â' ò B 11 B) 11 : 14
C) 22 : 7 D) 1 : 1
- Q.27.** 'Y?'%>@ "% 'iM'ù> 'YK 56 "%G 'Ù>'r &"y("r * "6"y7"12 29 '©M" >'©M'B 9"1\$"â 9"%d •™8" 8•) "Ù/"â "2 , 8"r -"é 'iG'%G '©0 "iG
A) 5 B) 3
C) 7 D) 1
- Q.28.** 1500 'Y@ 'y(" >"i? '©0 2 "Y0"Ù7 'Y> "%>'y>" # 'ÉM'ù>Â "C 9"%d "Y>" M"y?'R , "Ù/"é 'lO 'YM'ù> "™H?
A) 4% B) 5%
C) 6% D) 3.5%
- Q.29.** •ù 'É2"Ù2"y,"é '%G •Y*"%>@ 11"Y@•" *é0" .y 100 " ('É("é 'É?"%8"r "% "â "%\$ 5 " ('É""Â 'ù>-B 5" '©>" @ 'YG 'É>'b
A) 50 " (/ 50 Runs B) 45 " (/ 45 Runs
C) 55 " (/ 55 Runs D) 60 " (/ 60 Runs
- Q.30.** 9 'lO 25 'Y> 'é""Ù/"é(" *é\$" „ÖV â &÷ ÷ tional) 'YM'ù> "™K'y>?
A) 15 B) 17
C) 12 D) 20

- Q.31.** A 'Y?'%@ 'Y>' M'd '2 ' &"ù('1 'éG•" " " %@@ 'Y>' M'd '2 , &"ù('1 'éG•" *)0"â " 8'Y\$"â 9"%d "YG 4 'i?'%K•" \$'R 'R 8"é%
 A) 6 'i?'€ B) 8 'i?', f, F ys)
 C) 10 'i?'€ D) 7 'i?'€
- Q.32.** 180 'é@'ù0 ") 'É@ •ù " G") "é!"É@ 54 'Y?'é@/'% 'ù> 'Y@ 'y\$"ò 8'r "" 0"TM@ "TMH-B 'R •)-"r '2 *"é0 'Y0"%G 'éG•" /'" "ù\$'%
 A) 10 "%G'Y ' B) 12 "%G'Y ' f " 6V6öæG2•
 C) 15 "%G'Y ' D) 18 "%G'Y ' "
- Q.33.** '15,000 'Y@ 'y(" >"i? '©0 2 "Y0"Û7 'YG ")?•ò R 5"é0"Û7"ù 'i0 "%G '© "Û0"YC'iM'y? 'ÉM'ù>'Â " 8"é"é0'2 , "Û/"é 'Y> •Y 'I0 'Y
 A) '120 B) '150 ('150)
 C) '180 D) '200
- Q.34.** •ù 'iA'Y>'%&"é0 'Y?'%@@ "Y8"Û\$" "r •) "ù\$ 'éB")M'd *" # R " ") 'iG'%G 'YG 'É>'b -" # R "â 2"é- •Y0"Û "ù\$ 'Y0'I> "TMH-B /
 A) '650 B) '700
 C) '750 ('750) D) '800
- Q.35.** •ù 'YH")G 'éG•" " Â S *"%8"r " #R *"%8"r "r 8"ù "Û"r 2 4 : 5 'YG •Y(" *"é\$ 'éG•" 9"% -B 'i? 'YH")G 'éG•" " 2 'y(" >"i? '170
 A) 80 B) 100
 C) 120 (120 coins) D) 150
- Q.36.** •ù ") 'Â 5"9\$"Û\$" / 'ÉG") ('YG •i"é0 'Y@ 'iM" ?'ÉM'ù> 7 "%G'é@ 'I0 •© '©>•, 8"y." 9"%d •y8'Y> 'YA"" *"97"Û " / 'YM"yG'iM" •
 A) 648 "%G'é@² B) 748 "%G'é@² (748 cm²)
 C) 848 "%G'é@² D) 924 "%G'é@²
- Q.37.** 'ù&"ò 6-â , + cos ;, Ò !£" 6÷2 , 'IK cos ;, Ò 6-â , 'Y> 'é>', "Û/"â 9"1 "ãð
 A) " 2 sin ;€ B) " 2 tan ;€
 C) sin ;€ D) 1
- Q.38.** 'ù&"ò 8'ÉA'©M'©/ A '©0 "% 'É 'r " 8"Û5'IA")M'd ...&Vf/ÆPxive), "%."é?'B ...7-ÖÖWG ic) 'I%"â 8•) "Û0"é.'R ...@ransitive) 'I@'%
 A) " ?'YM'B 8•,•)' / Empty Relation B) 'IA")M'ù\$"â 8•,•)' / Equivalence Relation
 C) '©M" \$"ù8'É."ù\$ "% 'É 'r ò çF"×7-ÖÖWG ic Relation D) "%>" M"Y\$"Û0"ù "% 'É 'r ò Væ—`ersal Relation
- Q.39.** 'ù&"ò "+"(f: R !' R, f(x) = 2x 'iM"Y>" > '©0"ù-"é7"ù\$ "TMH, 'IK f 'Y?" , *"Û0'Y>" "â "+"("TMH?
 A) •ù "% " \$'Y> •i "Û "é&'R ò öæRÖöæR æB öçFò „&-|V7F—`e)B) 'É9" Û 'R \$'Y> •i "Û "é&'R ò Ö ày-one and Onto
 C) •ù "% " *" 'IA •Y("é "Û "é&'R ò öæRÖöæR ut not Onto D) ', \$"2 'YH'Y@ ', 9" '©M'1>'i / Neither One-one nor Onto
- Q.40.** 'éA'iM'ò ."é("i>'i> (Principal Value Branch) 'éG•" 6-â »'f ó" "â ."é('YM'ù> "TMK'y>?
 A) < ób f3 • B) < ó2 fç •
 C) < ób fC[• D) < ó" f" •
- Q.41.** tan {1(1) + cos {1(-1/2) + sin {1(-1/2) 'Y> 'é>', "Û/"â 9"1 "ãð
 A) < ó@ B) < ó
 C) 3< ó@ D) <
- Q.42.** sin {1(sin(2< ó2" "â 8"TM@ 'é>', "Û/"â 9"1 "ãð
 A) 2< ó0 B) < ó0
 C) -< ó0 D) 4< ó0
- Q.43.** 'ù&"ò "YM'ùB"" Ò ¶ ö-¥Ò 'R 8'É."ù\$ •i5"Û"/)9 (Symmetric Matrix) "TMH, 'IK 'YL', 8" 6" M'B 8'iM'ò 9"1 " ð
 A) a_ij = -a_ji B) a_ij = a_ji
 C) a_ij = 0 D) a_ij = 1
- Q.44.** 'ù&"ò " " 8'é>', "1 "ò "r 5" M'r "YM'ùB"" 9"% , 'IK (AB)' 'Y> 'é>', "ù8'YG 'É0"é," 9"1\$"â 9" fð
 A) A'B' B) B'A'
 C) AB D) BA
- Q.45.** 'ù&"ò 'R 2 r 2 "1 "ò "â 5"Û/" \$"Û "Û0'é#" / (Invertible) •i5"Û"/)9 "TMH, 'IK |adj(A)| 'Y> 'é>', "ù8'YG 'É0"é," 9"1 "ãð
 A) |A| B) |A|²
 C) |A|³ D) 3|A|
- Q.46.** 'ù&"ò "ù8" 8"é0'9?'R " "1 'iK '© 'YM'I?'ù>• ...&ðws) •Y%"Y> 'iK "%M'I 'Ò „6öÇVÖç2' 8" M"Y8'â „-FVçF-6 Â' 9"1 , 'IK "%>" #
 A) 1 B) 0 / Zero
 C) -1 D) •Y*" ?'é?'B ò -æf-æ—FP

- Q.47.** •ù “Y0”Ù •i5”Ù/”)9 A “YM’ùA’IM’YM” .’9@’ò „-àvertible) ‘Y9“>’I> “TMH ’ù&“ò “ ”y5”” /i?:
 A) $|A| = 0$ B) $|A| \neq 0$ (•Y5”Ù/” \$”Ù ”Ù0’é#” / ’%9” , Non-singular)
 C) $A = A'$ D) $A = -A'$
- Q.48.** ’12, ’ Ò ÆÖr†6-â , ’ “â , ”r 8“é*y ”Ù7 •Y5’Y2’Â †G’öG, ’ ”Ù/“â 9”1 “ãð
 A) $\tan x$ B) $\cot x$
 C) $-\cot x$ D) $\sec x$
- Q.49.** ’ù&“ò ’ Ò F â »’,f7, Ò <2’ ò f Ò 7<”” 9”2Â \$”2 G’öG, “â .“é(‘YM’ù> “TMK’y>?
 A) $3 / (1 + x^2)$ B) $1 / (1 + x^2)$
 C) $3 / (1 - x^2)$ D) $2 / (1 + x^2)$
- Q.50.** ’ù&“ò , Ò 6÷2 , ’l%“â ’ Ò 6-â , “TMK, ’IK dy/dx ‘Y> ’é>’, ”Ù/“â 9”1 “ãð
 A) $\tan ;\text{€}$ B) $-\cot ;\text{€}$
 C) $\cot ;\text{€}$ D) $-\tan ;\text{€}$
- Q.51.** “YC’IM’B ”r ”Ù7”y\$”Ù0’12 ’éG•” \$”Ù0”ù ”Ù/“â ” ”r 8“é*y ”Ù7 ’©0”ù5“ M’l(‘Y@ ’i0 ‘YM’ù> “TMK’y@ ’É, r = 5 cm “TMK?
 A) $10 < 6\text{Ü}''\text{ö6Đ}$ B) $5 < 6\text{Ü}''\text{ö6Đ}$
 C) $25 < 6\text{Ü}''\text{ö6Đ}$ D) $20 < 6\text{Ü}''\text{ö6Đ}$
- Q.52.** “Y ”Ù0 y = $x^3 - x$ ‘YG ’É?•)&” , Ò ” *” 8”Ü*” M“b 0”y “â ” *”Ü0“Y#’I> (Slope of Tangent) ‘YM’ù> “TMK’y@?
 A) 11 B) 12
 C) 10 D) 6
- Q.53.** ”+ $(1 / (1 + x^2)) dx$ ‘Y> ’é>’, ”Ù/“â 9”1\$“â 9”fð
 A) $\sin \{^1 x + C$ B) $\tan \{^1 x + C$
 C) $\log(1 + x^2) + C$ D) $\cos \{^1 x + C$
- Q.54.** ”+ $\sec x dx$ ‘Y> “%.”é “)(’é>’, ”Ù/“â 9”1 “ãð
 A) $\log |\sec x + \tan x| + C$ B) $\log |\sec x - \tan x| + C$
 C) $\sec x \tan x + C$ D) $\tan^2 x + C$
- Q.55.** ”+ e.2 †6-â , ^2 6÷2 , ’ G, “â .“é(‘YM’ù> “TMK’y>?
 A) e.2 6÷2 , ^2 0 B) e.2 6-â , ^2 0
 C) -e.2 6-â , ^2 0 D) e.2 †6-â , Ò 6÷2 , ’ ^2 0
- Q.56.** ’%?”iM’©?’B 8’é>’Y2’, ”2 âfÀ/2) (sin x / (sin x + cos x)) dx ‘Y> ’é>’, ”Ù/“â 9”1 “ãð
 A) < ó B) < ó@
 C) 1 D) 0
- Q.57.** ’ù&“ò b†,’ ’R 5“ù7’â +“((Odd Function, $f(-x) = -f(x)$) “TMH, ’IK ”+ < • C f(x) dx ‘Y> ’é>’, ”Ù/“â 9”1 “ãð
 A) 2 ”+ € C f(x) dx B) 0 / Zero
 C) a D) 1
- Q.58.** •Y5’Y2 “%.” “ # $(d^2y/dx^2)^3 + (dy/dx)^2 + \sin(dy/dx) + 1 = 0$ ‘Y@ ’%>’B „FVpree) ‘YM’ù> “TMH?
 A) 3 B) 2
 C) 1 D) ’©0“ù-“é7“ù\$ ’%9” / Not Defined
- Q.59.** “ H’i?’R “Y ”” 8’é@‘Y0’2 G’öG, ^2 ’ Ò “â 8’é>’Y2’, ” #’R „-çFVprating Factor - I.F.) ‘YM’ù> “TMK’I> “TMH?
 A) $e^{(+ P dx)}$ B) $e^{(+ Q dx)}$
 C) $+ P dx$ D) $e^{(- + P dx)}$
- Q.60.** •Y5’Y2 “%.” “ # $dy/dx = y/x$ ‘Y> “YM’ù>’© “TM2 (General Solution) ‘YM’ù> “TMK’y>?
 A) $y = Cx$ B) $y = C / x$
 C) $y = x^2 + C$ D) $xy = C$
- Q.61.** ’ù&“ò &”2 6”)(”Ù/”y\$”Ù\$” 8’i?’iK•” r “ ”r ”r 2”ù a × . b x = 0 “TMK, ’IK •TM(‘YG ’É@’¢ “â ”1# , , ”Ù/“â 9”1 “ãð
 A) 0° B) 45°
 C) 90° (< ó”Â *” 8”Ü*” 2•), “Y\$ / Perpendicular) D) 180°
- Q.62.** i0” r †E × k0”” “â .“é(‘YM’ù> “TMK’y>?
 A) 0 B) 1
 C) -1 D) k0

Q.63. $\frac{a}{x} = 2i0^2 \cdot 6E + 6k0$ "â * " ?'é>'2 „Ö væ—GVFR Æ wÂ' "ù\$'‰> “™K'y>?

- A) 7
B) 11
C) 49
D) " 41

Q.64. 'ù&"ò r r " r Ò r 9"2Â \$"2 8'i?"b r " " r "r 8•),)' 'éG•" "Ù/"â 8'IM"ò 9"f ð

- A) $a \times '10 b \times '©0‰M'©0$ " 'É5'B 9"‰ / Perpendicular
B) $a \times '10 b \times '©0‰M'©0$ "‰. "é("é '10 / "‰ " G'b 9"‰ (Parallel / Collinear)
C) $|a \times| = |b \times|$
D) $a \times + b \times = 0 \times$

Q.65. 'ù&"ò 'R 0"y "â " &"ù "ÒÙ "1 "Ù/"é • „F—&V7F—ôâ 6÷6—æW2' ÂÂ ÒÂ â 9"1 , 'IK 'YL', 8"â 8•),)' "‰&"‰5 "‰\$"Ù/ "™H?

- A) $l + m + n = 1$
B) $l^2 + m^2 + n^2 = 1$
C) $l^2 + m^2 + n^2 = 0$
D) $l^2 - m^2 + n^2 = 1$

Q.66. 'É?•)& " ‡,Â •, z) 'Y@ 'éB"" , "ù 'iA (Origin - 0, 0, 0) "‰G 'iB@ @ 'YM'ù> “™K'y@?

- A) $x + y + z$
B) " $(x^2 + y^2 + z^2)$
C) $x^2 + y^2 + z^2$
D) " $(x + y + z)$

Q.67. 'IK " G'i>•ù 'É?'‰ "r &"ù "ÒÙ "‰A'©>'B "Ù0'é6•2 Â " Â 2 \$'Y> a „, b „, c , "™H•"Â * 8"Ù* 2•), "Y\$ (Perpendicular) "™K•) "

- A) $a \cdot a , = b \cdot b , = c \cdot c ,$
B) $a \cdot a , + b \cdot b , + c \cdot c , = 0$
C) $a \cdot a , + b \cdot b , + c \cdot c , = 1$
D) $a \cdot b , - a \cdot b \cdot c = 0$

Q.68. " H'i?'R * "Ù0"1 "Ù0"é. , 8'é8"Ù/"â „Â ' . "y •™&"Ù&"y6"Ù/ '12' „ö&V7F—`e Function) Z = ax + by 'Y> •y7"Ù 'I. 'é> , „÷ F—Ö Â Val

- A) "‰A"‰ 'y\$ 'YM"yG'IM" "r 6" 0"Ù7 ('YK'9@'ò' , "ù 'iA'9 '©0 / Center of Feasible Region Only at origin
C) "‰A"‰ 'y\$ 'YM"yG'IM" "r " 'YG•)&"Ù0 '©0 / Exact center
D) 'YM"yG'IM" "r "ù8" - " / "é&"9 "Ù "ù 'É?•)& " * "ò ày random point

Q.69. 'ù&"ò " " &"2 8"Ù5'I 'IM" 'ù("é • „—æFW VæFVçB W`ents) "™K•"Â \$"2 „ " " " "â . "é('YM'ù> “™K'y>?

- A) $P(A) + P(B)$
B) $P(A) \times P(B)$
C) $P(A) / P(B)$
D) $P(A) - P(B)$

Q.70. 'ù&"ò „ ' Ò 2ó,Â „ " Ò ó" \$'Y> P(A ") B) = 1/4 "™K, 'IK P(A | B) 'Y> 'é> , "Ù/"â 9"1 "ãð

- A) 1/2
B) 2/3
C) 3/4
D) 1/8

Q.71. 'Y?'‰@ '‰ '‰> E 'Y@ '©M" >'ù?'Y\$"â „R' \$'Y> •™8'Y@ '©B" '‰ '‰> P(E) 'Y> 'ùK'r "Ù/"â 9"1\$ "â 9"f ð

- A) 0
B) 1
C) 0.5
D) •Y("ù6"Ù "ù\$ / Indeterminate

Q.72. 'ù&"ò 'R 5" M'r "YM'ùB"" 9",Â \$"2 2 r 8'iH"R "ù8 '©M" "é0 'Y> •i5"Ù/"9 "™K'I> "™H?

- A) "Y?"y. - "‰. "é?'B "YM'ùB"" ò 6°ew-symmetric
B) "‰. "é?'B "YM'ùB"" ò 7—ÖÖWG ic Matrix
C) •y "é •i5"Ù/"9 / Identity Matrix
D) "iB"‰M'ò "YM'ùB"" ò çVÆÂ Ö G ix

Q.73. '12', b‡, ' Ò <2 Ò 7, "â 8"Ù‰"é(" / •™ "Ù "ù7"Ù (Local Maxima) 'Y? , "ù 'iA '©0 "™K'y>?

- A) $x = 1$
B) $x = -1$
C) $x = 0$
D) $x = 3$

Q.74. "+ (1 / " (a² - x²)) dx 'Y> 'é> , "Ù/"â 9"1\$ "â 9"f ð

- A) $\sin \{^1(x/a) + C$
B) $(1/a) \sin \{^1(x/a) + C$
C) $\cos \{^1(x/a) + C$
D) $\log|x + " (a^2 - x^2)| + C$

Q.75. 'i' 6•2 8'é>'Y2' , „—çFVpration by Parts) "‰B'IM" "2 R b G, Ò R "2 b G, Ò "2 •Rr , "2 b G, •Ò G , . "y '©M" %'â +")('©A'‰("r "â 8"™

- A) BODMAS '‰?'ù.
B) ILATE '‰?'ù. (Inverse, Logarithmic, Algebraic, Trigonometric, Exponential)
C) L'Hopital '‰?'ù.
D) Cramer '‰?'ù.

Q.76. $2 \tan \{^1 x 'Y> 'é> , 6-â \gg "r *iK•" . "y 'YM'ù> "™K'I> "™H?$

- A) $\sin \{^1(2x / (1 + x^2))$
B) $\sin \{^1(2x / (1 - x^2))$
C) $\sin \{^1((1 - x^2) / (1 + x^2))$
D) $\sin \{^1(x / (1 + x^2))$

Q.77. 'ù&"ò "YM'ùB"" " "1 "ò " r 2 \$'Y> B 'Y@ 'YK'ù? 3×4 "™K, 'IK 'yA'9('12 •i5"Ù/"9 AB 'Y@ 'YK'ù? (Order) 'YM'ù> “™K'y@?

- A) 3×3
B) 2×4
C) 4×2
D) 'yA'9("‰ 'Ù5 '‰9" "™H / Multiplication not possible

Q.78. x-•Y "Ù7 'Y@ 'i?'YM-'YK'ÉM"ù>•ù (Direction Cosines of x-axis) 'YM'ù> "™H•#ð

- A) (1, 0, 0)
B) (0, 1, 0)
C) (0, 0, 1)
D) (1, 1, 1)

- Q.79.** $f(x) = \begin{cases} x^2 + 1 & \text{if } x \leq 0 \\ x^2 + 2x + 1 & \text{if } x > 0 \end{cases}$ Is $f(x)$ continuous and differentiable at $x = 0$?
- A) Yes, both continuous and differentiable
B) Yes, continuous but not differentiable
C) No, neither continuous nor differentiable
D) No, continuous but not differentiable
- Q.80.** $y = \frac{1}{1+x^2}$. Find $\frac{dy}{dx}$ at $x = 1$.
- A) $-\frac{1}{2}$
B) $-\frac{1}{4}$
C) $-\frac{1}{8}$
D) $-\frac{1}{16}$
- Q.81.** $y = 3x^2 + 4x - 5$. Find $\frac{dy}{dx}$ at $x = 2$.
- A) 12
B) 16
C) 7
D) 0
- Q.82.** $y = \sin(x)$. Find $\frac{dy}{dx}$ at $x = \frac{\pi}{4}$.
- A) $\frac{1}{\sqrt{2}}$
B) $\frac{1}{2}$
C) $\frac{1}{4}$
D) $\frac{1}{8}$
- Q.83.** $y = \cos(x)$. Find $\frac{dy}{dx}$ at $x = \frac{\pi}{2}$.
- A) 75
B) 80
C) 90
D) 100
- Q.84.** $y = \tan(x)$. Find $\frac{dy}{dx}$ at $x = \frac{\pi}{4}$.
- A) $2^2 \times 5 \times 7$
B) $2 \times 5^2 \times 7$
C) $2^3 \times 5 \times 7$
D) $2 \times 5 \times 7^2$
- Q.85.** $y = \sec(x)$. Find $\frac{dy}{dx}$ at $x = \frac{\pi}{4}$.
- A) 5
B) 9
C) 16
D) $\frac{22}{7}$
- Q.86.** $y = \csc(x)$. Find $\frac{dy}{dx}$ at $x = \frac{\pi}{4}$.
- A) $-\frac{1}{\sin(x)}$ (Terminating)
B) $-\frac{1}{\sin(x)}$ (Non-terminating)
C) $-\frac{1}{\sin(x)}$ (Non-terminating)
D) $-\frac{1}{\sin(x)}$ (Non-terminating)
- Q.87.** $y = \cot(x)$. Find $\frac{dy}{dx}$ at $x = \frac{\pi}{4}$.
- A) $-\frac{1}{\tan(x)}$
B) $-\frac{1}{\tan(x)}$
C) $-\frac{1}{\tan(x)}$
D) $-\frac{1}{\tan(x)}$
- Q.88.** $y = \operatorname{cosec}(x)$. Find $\frac{dy}{dx}$ at $x = \frac{\pi}{4}$.
- A) $x^2 - 4x + 1$
B) $x^2 + 4x + 1$
C) $x^2 - 4x - 1$
D) $x^2 + 4x - 1$
- Q.89.** $y = \sec(x)$. Find $\frac{dy}{dx}$ at $x = \frac{\pi}{4}$.
- A) $a \cdot a, b \cdot b, c \cdot c$
B) $a \cdot a, b \cdot b, c \cdot c$
C) $a \cdot a, b \cdot b, c \cdot c$
D) $a \cdot a, b \cdot b, c \cdot c$
- Q.90.** $y = \csc(x)$. Find $\frac{dy}{dx}$ at $x = \frac{\pi}{4}$.
- A) $\frac{1}{\sin(x)}$ (Infinitely many)
B) $\frac{1}{\sin(x)}$ (Infinitely many)
C) $\frac{1}{\sin(x)}$ (Exactly One Unique Solution)
D) $\frac{1}{\sin(x)}$ (Two Distinct Solutions)
- Q.91.** $y = \cot(x)$. Find $\frac{dy}{dx}$ at $x = \frac{\pi}{4}$.
- A) 6
B) 3
C) 2
D) 4
- Q.92.** $y = \operatorname{cosec}(x)$. Find $\frac{dy}{dx}$ at $x = \frac{\pi}{4}$.
- A) $b^2 - 4ac$
B) $b^2 + 4ac$
C) $4ac - b^2$
D) $(b^2 - 4ac)$
- Q.93.** $y = \sec(x)$. Find $\frac{dy}{dx}$ at $x = \frac{\pi}{4}$.
- A) $\frac{1}{\sin(x)}$ (Real and Equal Roots)
B) $\frac{1}{\sin(x)}$ (Real and Equal Roots)
C) $\frac{1}{\sin(x)}$ (Real and Equal Roots)
D) $\frac{1}{\sin(x)}$ (Real and Equal Roots)
- Q.94.** $y = \csc(x)$. Find $\frac{dy}{dx}$ at $x = \frac{\pi}{4}$.
- A) ± 4
B) ± 2
C) ± 8
D) 0

Q.95. A.P. 3, 8, 13, 18, ... 'Y> 10"Y>• *'b "Ù/"â 9"1 "ãð

- A) 48 B) 50
C) 45 D) 53

Q.96. '©M" %'â ",*)0"Ù#"é 'YK•" f 2 " 2 2 2 âââ + 100) 'Y> 'ùK'y+" "Ù/"â 9"1 "ãð

- A) 5050 B) 5000
C) 5150 D) 5500

Q.97. A.P. 7, 13, 19, ..., 205 'éG•" " 2 'Y?'I("r *'b 9"%o ?

- A) 34 B) 33
C) 35 D) 32

Q.98. %³ABC 'éG•" DR "R \$2 9"%od 'ù&"ò B Ò ãR 6ÒÂ D" Ò 2 6ÒÂ R Ò 6Ò 9"2Â \$"2 T2 "â ."é('YM'ù> "TMK'y>?

- A) 2 cm B) 1.5 cm
C) 2.5 cm D) 3 cm

Q.99. 'iK "%o." B'¢ \$"Ù0"ù-" "1 'Y@ 'ÙA'É>'9 'Y>•Y(" **é\$ 4 : 9 "TMH-B "%o "r "Ù7"y\$"Ù0'12"1 'Y>•Y(" **é\$ 'YM'ù> "TMK'y>?

- A) 16 : 81 B) 2 : 3
C) 4 : 9 D) 81 : 16

Q.100. •ù "%o.'YK'2 \$"Ù0"ù-" 'Y> 'Y0"Ù# 13 "%oG'é@ 'I0 •ù 'ÙA'É> 5 "%oG'é@ "TMH-B \$" 8" @ 'ÙA'É> 'Y@ ")."Ù,"é 'YM'ù> "TMK'y>?

- A) 12 cm B) 10 cm
C) 11 cm D) 8 cm

Q.101. 'É?'%oM'iA'9 (2, 3) 'I0 (4, 1) 'YG 'É@'¢ " &")0" "Ù/"â 9"1 " ð

- A) 2" 2 'é>'IM" / 2" 2 units B) " 8 'é>'IM" / " 8 units
C) 4 'é>'IM" / 4 units D) 2 'é>'IM" / 2 units

Q.102. 'É?'%oM'iA' (-6, 8) 'Y@ 'éB"), "ù("Ù&" „÷ igin) "%oG 'iB" @ 'YM'ù> "TMK'y@?

- A) 10 'é>'IM" / 10 units B) 14 'é>'IM" / 14 units
C) 8 'é>'IM" / 8 units D) 6 'é>'IM" / 6 units

Q.103. 'É?'%oM'iA'9 (-5, 7) 'I0 (-1, 3) 'YK 'é?">"%oG "Y>")G " G'í>'i ' "r .yM'òÙ, "ù("Ù&" "r ("ù0"Ù&"y6"é 'R "Ù/"â 9"1 'yG?

- A) (-3, 5) B) (-2, 5)
C) (3, -5) D) (-3, -5)

Q.104. $\sin 60^\circ \times \cos 30^\circ + \cos 60^\circ \times \sin 30^\circ$ 'Y> 'é>', "Ù/"â 9"1 "ãð

- A) 1 B) 1/2
C) " 3/2 D) 0

Q.105. $9 \sec^2 A - 9 \tan^2 A$ 'Y> 'é>', "ù8'YG 'É0"é," 9"fð

- A) 9 B) 1
C) 8 D) 0

Q.106. $\tan 65^\circ / \cot 25^\circ$ 'Y> 'é>', "Ù/"â 9"1 "ãð

- A) 1 B) 0
C) " 3 D) 1/" 3

Q.107. •ù •©0"Ù""Ù5"é"" •)-"r " * "é •TM8'Y@ •© '©>•, "r, " >'É0 "TMH-B 8")0"Ù/ 'Y>•TM("Ù('ù('YK'2 „ ævÆR öb VÆPvation) 'YM

- A) 45° B) 30°
C) 60° D) 90°

Q.108. 'ÙB'é? 'YG •ù 'É?•)&" 8"r ." ("é0 'YG '©>'b 8"r 3 ." " &")0" * " ("é0 'YG "i?'i0 'Y>•TM("Ù('ù('YK'2 3 9"%od 'é@'%">" " •)

- A) 10" 3 m B) 30" 3 m
C) 15 m D) 20 m

Q.109. "YC'IM'B "r "ù8", "é9"Ù/ 'É?•)&" 8"r " '©@ 'y 'iK "%oM'©0"Ù6 " G'í>'9 PA 'I0 PB 'Y@ ")."Ù,"é 'ùK•". "y 'YM'ù> "%o 'É' r 9"1

- A) PA = PB ("%"é("TMK'I@ "TMH•" B) PA > PB
C) PA < PB D) PA = 2PB

Q.110. •Y0"Ù""YC'IM'B .y 'É("â "1# (Angle in a semi-circle) 'Y?'I("â 9"1\$"â 9"fð

- A) "%o.'YK'2 f" ' ò &-v†B ævÆR f" • B) 60°
C) 180° D) 45°

- Q.111.** 'ù&"ð 8"Û*" M"b 0"y "â A 'lO PB •ù 'iB"%0"r 8"r f "r "1# '©0 'ÙA'Y@ "™K•"Â \$"2 " ðA 'Y> 'é>', "Ù/"â 9"1 "â „ð "y 'iM" 9", "ð
A) 50° B) 60°
C) 70° D) 80°
- Q.112.** 'iM" ?'ÉM'ù> r "Y>")G "YC'iM'B "r " , \$"Û0"ù "Ù/i ' "â "Û7"y\$"Û0'12 'YM'ù> "™K'y> 'É?"% "â "1# ;, 9" f ð
A) (;, ò 3c ' r Âr² B) (;, ò f ' r Âr²
C) (;, ò 3c ' r #Âr D) 2< +
- Q.113.** •ù "YC'iM'B " * " ?y? 'lO 'YM"yG'iM" +"" 8•) "Ù/"é\$"Û. 'R 0")* "%oG 'É0'é, " 9"%o –B " , 5"9\$"Û\$ 'Y@ 'iM" ?'ÉM'ù> 'YM'ù> "™K
A) 2 'é>'iM" / 2 units B) < .é\$"Û0'R ò Â units
C) 4 'é>'iM" / 4 units D) 7 'é>'iM" / 7 units
- Q.114.** 'iM" ?'ÉM'ù> r "Y>")G •ù ' K", " M'y "12"r "â 8'éM'©B" M'2 *97"Û " / 'YM"yG'iM" +"" "Ù/"â 9"1\$"â 9" f ð
A) 3< + B) 2< +
C) 4< + D) 2/3 < +0
- Q.115.** 'ù&"ð 'R "12"r " \$"Û0"ù "Ù/"â &"1 " (" " &" "é , 'lK •™8'Y> •i/'l('Y?'l("r " ("â 9"2 "é 'y>?
A) 8 'yA'%o> / 8 times B) 4 'yA'%o> / 4 times
C) 2 'yA'%o> / 2 times D) 16 'yA'%o> / 16 times
- Q.116.** "%o."é('iM" ?'ÉM'ù> 'lO "%o."é(•© '©>•, 5"é2"r 'R , "y2', " 'R 6•) " "r 'ù\$"‰K•" "â "%oA'©>'B "Ù/"â 9"1 "ã ð
A) 3 : 1 B) 1 : 3
C) 1 : 1 D) 2 : 1
- Q.117.** 'é>'yM'ðÂ .é"Û/'R " , "™A") 'YG 'É@'¢ 8"™@ •i(" -"Y?'R 8•), •) (Empirical Relationship) 'YM'ù> "™H?
A) 'É9" 2'R Ò 2 .é"Û/'R Ò " .é"Û/ / Mode = 3 Median - 2 Mean B) 'É9" 2'R Ò " .é"Û/'R Ò 2 .é"Û/ / Mode = 2 Median - 3 Mean
C) 3 'É9" 2'R Ò .é"Û/'R Ò " .é"Û/ D) 'é>'yM'ù = 3 'É9" 2'R Ò " .é"Û/
- Q.118.** "%o 'iM'ù>9 2, 4, 6, 8, 10 'Y> "%o."é 'lO 'é>'yM'ð "Ù/"â 9"1 "ã ð
A) 6 B) 5
C) 7 D) 8
- Q.119.** •i 'Y!'ÉK•" rÂ "Â Â BÂ 'Â RÂ , " .é"Û/"ù "â „ÖVF– â" "Ù/"â 9"1 " ð
A) 7 B) 8
C) 6 D) 5
- Q.120.** 'É 'ù(2, 3, 5, 3, 7, 3, 8, 3, 9 'Y> 'É9" 2'R "Ù/"â 9"1 "ã ð
A) 3 B) 5
C) 7 D) 9
- Q.121.** 'Y?"%o@ •Y8•)-"R 'ù("â „–x ÷76– le Event) 'Y@ '©M" >ù?'Y\$"â "Ù/"â 9"1\$" 9" f ð
A) 0 B) 1
C) 0.5 D) -1
- Q.122.** •ù '©>%oG 'YK •ù 'É>" +y 'Y("r * " 'Ù>'ÉM'ð 8•) "Ù/"â ... ime Number) •i("r " * "Û0"é/"ù 'l> 'YM'ù> "™K'y@?
A) 1 / 2 B) 1 / 3
C) 1 / 6 D) 2 / 3
- Q.123.** •Y "Ù " \$" 9 '1G•) " •, S" *iM'lK•" " \$"é6 'Y@ 'y!"Ù!" .y "%oG •ù '©\$"Û\$â ("ù "é2"â "é\$"â 9"%o d •ù ")>"" 0•) 'Y> 'É>i6"é9
A) 1 / 26 B) 1 / 52
C) 1 / 13 D) 2 / 13
- Q.124.** •ù 'Y>" 8"r " \$"R c "ù." ù •) "â 8"r "é\$" 9", \$"Y> 40 'Y?'é@/'%o 'ù> "%oG ")L'ù\$" 9"%o d '©B" @ 'ù>'iM" > 'Y@ 'l8'B "é2 'YM'
A) 48 'Y?'é@/'%o 'ù> B) 50 'Y?'é@/'%o 'ù>
C) 45 'Y?'é@/'%o 'ù> D) 52 'Y?'é@/'%o 'ù>
- Q.125.** 15,000 'Y@ " >"i? '©0 10% "Y>" M"y?'R &" 8"r " 5" M"r "r 4" " 4' "â •) \$" "Ù/"â 9"1 "ã ð
A) 150 B) 125
C) 1100 D) 175
- Q.126.** A 'Y?"%o@ 'Y>" M'ð "2 " &"ù('éG•" \$"Y> B 18 'i?', .y '©B" > 'Y0'l> "™H–B &"1("1 'é?) " "ù\$"‰G 'i?', .y '©B" > 'Y0"y 'yG?
A) 7.2 'i?', ò rÂ" F ys B) 8 'i?', ò , F ys
C) 6 'i?', ò b F ys D) 7.5 'i?', ò rÂ" F ys

- Q.127.** 'ù&"ð " (" "r.")2"Ù/ 'éG•" #RR " 5"9&"Ù""ð 9"2 "é, 'IK 'yC"™?9@ 'YK 'i*B."y 'Y?I("r *"Ù0'I?"i\$ 'Y@ 'Y." " (" 9"1 " \$"é "ð
A) 20% B) 25%
C) 16.66% D) 15%
- Q.128.** 'ùB'YM")?' 5"ù-"é ', *"'Ù0'ég'ù?'Y> a = bq + r 'éG•" 6"y7"12 r 'YG ")?•ò "É("%@ "i0"Ù\$ "%\$ "Ù/ "™H?
A) 0 "d r < b B) 0 < r "d b
C) 0 "d r "d b D) r > b
- Q.129.** "%,%G '1K'ù@ •Y-"é "Ù/ "% 'iM'ù> (Smallest Prime Number) 'YL', 8" 9"f ð
A) 2 (•ù 'é>'IM" 8'à 'Ù>'ÉM'ò' ò " „ðæÇ' W'en Prime) B) 1
C) 3 D) 0
- Q.130.** " 'H'i?'R , "™A'©& (Linear Polynomial) 'Y@ •Y""ù 'I. '%>'B "ù\$"%@ "™K'I@ "™H?
A) 1 B) 2
C) 0 D) 3
- Q.131.** 'ù&"ð &"Ù5"ù "é\$ 'É9" *b <" Ò 7, 2 " "r 6")("Ù/"é 'R ± 'I0," 9"1 , 'IK 1/; 2 ó2 'Y> 'é>', "Ù/"â 9"1 "ãð
A) 3 / 2 B) 2 / 3
C) -3 / 2 D) 1
- Q.132.** 'ù&"ð &"2 0"y "é •" 'R , "ù 'iA '©0 '©M" \$"ù "Ù "y& 'Y0'I@ "™H•"Â \$"2 5"" , "ù 'iA 'iK'%K•" 8'é@ 'Y0'9K•" "â "%8"â 9"" 9"1 "ãð
A) •Y&"Ù5"ù\$ / "™2 (Unique Solution) B) •Y(•)\$ •Y("y "™2 / Infinitely Many Solutions
C) 'YK•, 9"" ("™@•" ò æð 6ðÇWF-ðâ ... arallel Lines) D) 'iK "™2 / Two Distinct Solutions
- Q.133.** 'ù&"ð , 2 ' Ò B \$'Y> x - y = 4 "™K, 'IK x 'I0 y 'YG 'é>', "Ù/"â 9"1 'yG?
A) x = 9, y = 5 B) x = 8, y = 6
C) x = 10, y = 4 D) x = 7, y = 7
- Q.134.** 'iM'Y?'%>'B 8'é@ 'Y0'2 "â ."é('R 0")* (Standard Form) 'YM'ù> "™K'I> "™H?
A) $ax^2 + bx + c = 0$ ('É9"é a "" 0) B) $ax + b = 0$
C) $ax^3 + bx^2 + cx + d = 0$ D) $ax^2 + c = 0$
- Q.135.** "%." " # $x^2 - 9 = 0$ 'YG 'éB"" "Ù/"â 9"1 'yG?
A) ±3 B) 9
C) 3 D) ±9
- Q.136.** A.P. -10, -6, -2, 2, ... 'Y> "%>" M"R •)\$" †B' "Ù/"â 9"f ð
A) 4 B) -4
C) 2 D) -2
- Q.137.** '©M" %'â â 5"ù7'â *"'Ù0"é "9\$ "% 'iM'ù>'9 'Y> 'ùK'y+"" ...7VÒ öb f—'7B â öFB æ GW al numbers) 'YM'ù> "™K'I> "™H?
A) n^2 B) $n(n+1)$
C) $2n$ D) $n(n+1)/2$
- Q.138.** 'É?'%M'iA (-4, -5) 'Y?', 'IA" M'Y>•)6 'éG•" 8"Ù% "ù\$ "™H?
A) 'IC'I@'ò 'IA" M'Y>•)6 (Third Quadrant) B) '©M" %'â 'IA" M'Y>•)6 / First Quadrant (+, +)
C) 'iM'Y?'I@'ò 'IA" M'Y>•)6 / Second Quadrant (-, +) D) '©\$" 0"Ù% '©\$" 0"Ù%"é "b ò 4ourth Quadrant (+, -)
- Q.139.** 'ù&"ð \$" ('É?'%M'iA "% " G'i@ (Collinear) "™K•"Â \$"2 '‰8"r , "%("r 5"é2"r \$"Ù0"ù-" 'Y> 'YM'yG'IM" +"" "Ù/"â 9"1 "ãð
A) 0 B) 1
C) -1 D) •Y* " ?'é?'B 0")* "%G •Y("y / Infinitely Many
- Q.140.** 'IM" ?'ÙA'Â "r 6" 0"Ù7 (x •, y •), (x ,, y ,), (x f, y f) "™K•"Â \$"2 "y("Ù&"Ù0'R "r ("ù0"Ù&"y6"é 'R "Ù/"â 9"1 'yG?
A) $((x \bullet + x, +x f)/3, (y \bullet + y, +y f)/3)$ B) $((x \bullet + x,)/2, (y \bullet + y,)/2)$
C) $(x \bullet + x, +x f, y \bullet + y, +y f)$ D) $(x \bullet x, x f, y \bullet y, y f)$
- Q.141.** 'ù&"ð 6-â , = 3/5 "™K, 'IK cos ;, "â ."é('YM'ù> "™K'y>?
A) 4 / 5 B) 5 / 4
C) 3 / 4 D) 4 / 3
- Q.142.** tan 45° 'Y> 'é>', "Ù/"â 9"1\$ "â 9"f ð
A) 1 B) 0
C) 1/" 3 D) " 3

Q.143. $1 + \cot^2$; „Û8‘YG ‘É0“é,“ 9”1\$“â 9”fð

A) cosec^2 ;€

B) $\sec^2$;€

C) $\tan^2$;€

D) $\sin^2$;€

Q.144. “YC‘IM‘B”r ‘R,“ù ‘iA ‘©0 ‘Y?’I(“ 8”Û*“ M“b 0”y “é•” ” ‘©@ ‘É> “% ‘l@ “TMH•#ð

A) ‘YG“Y2 1 (Only one)

B) 2

C) •Y(•)\$ / Infinity (“)

D) 0

Q.145. •ù “YC‘IM‘B” ” “ù\$%@ “%.“é ‘l0 “%M‘©0”Û6 “ G‘i>•ù (Parallel Tangents) “TMK “% ‘l@ “TMH•#ð

A) 2

B) 4

C) 1

D) •Y(•)\$ / Infinity (“)

Q.146. ‘IM“ ?‘ÉM‘ù> r “Y>“)G “YC‘IM‘B”r ‘R ‘lA“ M‘Y>)6 (Quadrant) ‘Y> ‘YM“yG‘IM“ +““ ”Û/“â 9”1\$“â 9”fð

A) $1/4 < +$

B) $1/2 < +$

C) $< +$

D) $2 <$

Q.147. ‘IM“ ?‘ÉM‘ù>9 R ‘l0 r ($R > r$) “Y>“)G ‘iK “% ‘YG•)&”Û0” / “YC‘IM‘lK•” ”r, ” ‘É(“r 5“)/ (Ring) ‘Y> ‘YM“yG‘IM“ +““ ”Û/“â 9”fð

A) $< \dots +$ ” Ò +”•

B) $< \dots$ ” Ò ”>

C) $2 < \dots$ ” Ò ”•

D) $< +$ ” ò +

Q.148. ‘IM“ ?‘ÉM‘ù> 7 “%G‘é@ ‘l%“â •) “é 10 “%G‘é@ “Y>“)G “).”Û,“YC‘IM‘l@‘ò ,y2’, “â ‘ù\$, ”Û/“â 9”1 “ãð

A) $1540 \text{ “%G‘é@}^3 / 1540 \text{ cm}^3$

B) $1450 \text{ “%G‘é@}^3 / 1450 \text{ cm}^3$

C) $1500 \text{ “%G‘é@}^3 / 1500 \text{ cm}^3$

D) $770 \text{ “%G‘é@}^3 / 770 \text{ cm}^3$

Q.149. “i ‘YA ‘Y@ ‘l?“ M‘ù •© ‘©>•, ...6/Æ çB †V–v‡B vÂr’ “â 8”) \$”Û0 ‘YM‘ù> “TMH?

A) “ ($r^2 + h^2$)

B) $r^2 + h^2$

C) “ ($r^2 - h^2$)

D) $r + h$

Q.150. ‘ÙA‘É> a “Y>“)G •ù ‘%((Cube) ‘YG ‘éA‘iM‘ò 5“ù “ M’2 „F– vöæ Â’ ” 2‘ém‘É>•, ”Û/“â 9”1 ” ð

A) a” 3

B) a” 2

C) 3a

D) a / “ 3

Q.151. “Y0”Û •Y ‘l0“é2 10 - 25 ‘Y> “Y0”Û ‘©?“TM”, „6/Æ 72 Ö k) ‘YM‘ù> “TMK‘y>?

A) 17.5

B) 15

C) 20

D) 35

Q.152. •i ‘Yl“ÉK•” ”Â #RÂ RÂ ,Â 3RÂ # “â *” ?“%0 (Range) ‘YM‘ù> “TMK‘y>?

A) 23

B) 35

C) 12

D) 20

Q.153. ‘Y?“%@ ‘%?“iM‘©?’B ‘ù(“â ...7W&R W`ent) ‘Y@ ‘©M“ >‘ù?’Y\$“â “ù\$%@ “TMK‘l@ “TMH?

A) 1

B) 0

C) 0.5

D) •Y(“ù6”Û “ù\$ / Indeterminate

Q.154. ‘ù&“ð „R’ Ò ã R 9”2Â \$”2 tR (“TM@•”r ... †æ÷B R” ” *”Û0“é/“ù ‘l> ‘YM‘ù> “TMK‘y@?

A) 0.95

B) 0.05

C) 0.50

D) 1.05

Q.155. ‘iK “%?“YM‘YK•” ”2 ‘R 8“é% •TM “é2’%G ‘©0 ‘Y. “%G ‘Y. •ù ‘©?’IM‘B „ B ÆV 7B öæR †V B’ ‘%G ‘Y@ ‘©M“ >‘ù?’Y\$“â ”Û/“â

A) $3 / 4$

B) $1 / 4$

C) $1 / 2$

D) 1

Q.156. •ù “Y8”Û\$” ”2 “s# .”y ‘ÉG‘©(“r *” # R “â 2“é- “TMK‘l> “TMH–B 5“%M‘lA ‘Y> ‘YM“ / ‘éB“)M‘ò „6÷7B ice) ‘YM‘ù> “TMK‘y>?

A) 1600

B) 1580

C) 1640

D) 1560

Q.157. 120 ‘é@“ù0 “) ‘É@ ‘ùM“ G’, SB “ù.” ù •) “â ” “é2 “%G •ù ‘i ‘ÛG ‘YK ‘Y?’I(“r 8‘é/ ‘éG•” *”é0 ‘Y0”y ” ð

A) $8 \text{ “%G‘Y’ } \dot{} \text{ , } 6V6\ddot{}\text{æG0}$

B) $6 \text{ “%G‘Y’ } \dot{} \text{ b } 6V6\ddot{}\text{æG0}$

C) $10 \text{ “%G‘Y’ } \dot{} \text{ 6V6}\ddot{}\text{æG0}$

D) $12 \text{ “%G‘Y’ } \dot{} \text{ ” } 6V6\ddot{}\text{æG0}$

Q.158. (7 y u - 3 u x) ‘Y> •y “é ‘Y> •Y ‘R ”Û/“â 9”1 “ãð

A) 4

B) 6

C) 0

D) 7

- Q.159.** 'Y?'l(r 8'é/ 'éG•" "1 'y(5% "Y>" M"y?'R 8"é"é0'2 ,"ù/"é 'Y@ 'i0 "%G 'iK'yA'%"> "TMK 'É>•ù "ãð
A) 20 "Y0"Ù7 / 20 Years B) 10 "Y0"Ù7 / 10 Years
C) 15 "Y0"Ù7 / 15 Years D) 25 "Y0"Ù7 / 25 Years
- Q.160.** •ù "%. 'É>"MA 'IM" ?'ÀA'Â " *'Ù0'IM'ùG'R -" "â b 8"y. " 9"%d •y8'Y> 'YM"yG'IM" +"" "Ù/"â 9"1 "ãð
A) 9" 3 "%G'é@² / 9" 3 cm² B) 18" 3 "%G'é@² / 18" 3 cm²
C) 36 "%G'é@² / 36 cm² D) 6" 3 "%G'é@² / 6" 3 cm²
- Q.161.** 3/4, 5/6, 7/8, 9/10 'éG•" 8'É8"r, ' <" -"ù("Ù('YL', 8" 9"fð
A) 9 / 10 B) 7 / 8
C) 5 / 6 D) 3 / 4
- Q.162.** '©M" %'â S 8'â *'Ù0"é "9\$ "% 'iM'ù>'9 'Y> 'l8'B "Ù/"â 9"1 "ãð
A) 51 B) 50
C) 50.5 D) 100
- Q.163.** 'ù&"ð %â±, Ò ' 2 %â±, 2 ' Ò 3# 9"2Â \$"2, "â .é('YM'ù> "TMK'y>?
A) 7 B) 6
C) 8 D) 5
- Q.164.** 20% 'l0 10% 'Y@ 'iK 'YM" . "ù 'lB'ð ...7V66W76—`e Discounts) 'YG "%. 'lA")M'ð 'Y2 'lB'ð "Ù/"â 9"1 " ð
A) 28% B) 30%
C) 25% D) 22%
- Q.165.** "i>•)\$ 'É2 'éG•" ("é5 'Y@ '©>"" "ù." ù •) "â \$'Y> 'y>" > 'Y@ '©>"" " "ù." ù •) "â 9"%d 'y>" > 'Y@ 'i?'i> 'éG•" "é2 'YM'ù> "TMK
A) 12 'Y?'é@/'%" 'ù> B) 8 'Y?'é@/'%" 'ù>
C) 10 'Y?'é@/'%" 'ù> D) 14 'Y?'é@/'%" 'ù>
- Q.166.** '©>•y* A 'ù 'Y@ 'YK 10 "% 'ùG 'l%"â *é '¢ " R •) "r."y 'Ù0'l> "TMH-B &"1("1 'é?) " "ù\$'%G "%. 'ð ."y 'Ù0"y 'yG?
A) 6 "% 'ùG / 6 Hours B) 8 "% 'ùG / 8 Hours
C) 5 "% 'ùG / 5 Hours D) 7.5 "% 'ùG / 7.5 Hours
- Q.167.** 'YK•, "%0"é6"ð 8"é"é0'2 ,"ù/"é '©0 6 "Y0"Ù7"1 'éG•" 8"Ù5'ù 'Y@ 3 'yA'%"@ "TMK 'É>'l@ "TMH-B 5"é0"Ù7"ù 'ÉM'ù>'Â &" "l
33.3% 15.0%
12.5% 18.0%
- Q.168.** 'YK•, "%0"é6"ð 8"é"é0'2 ,"ù/"é '©0 7 "Y0"Ù7"1 'éG•" 8"Ù5'ù 'Y@ 3 'yA'%"@ "TMK 'É>'l@ "TMH-B 5"é0"Ù7"ù 'ÉM'ù>'Â &" "l
28.6% 15.0%
12.5% 18.0%
- Q.169.** 'YK•, "%0"é6"ð 8"é"é0'2 ,"ù/"é '©0 8 "Y0"Ù7"1 'éG•" 8"Ù5'ù 'Y@ 3 'yA'%"@ "TMK 'É>'l@ "TMH-B 5"é0"Ù7"ù 'ÉM'ù>'Â &" "l
25.0% 15.0%
12.5% 18.0%
- Q.170.** 'YK•, "%0"é6"ð 8"é"é0'2 ,"ù/"é '©0 9 "Y0"Ù7"1 'éG•" 8"Ù5'ù 'Y@ 3 'yA'%"@ "TMK 'É>'l@ "TMH-B 5"é0"Ù7"ù 'ÉM'ù>'Â &" "l
22.2% 15.0%
12.5% 18.0%
- Q.171.** 'YK•, "%0"é6"ð 8"é"é0'2 ,"ù/"é '©0 10 "Y0"Ù7"1 'éG•" 8"Ù5'ù 'Y@ 3 'yA'%"@ "TMK 'É>'l@ "TMH-B 5"é0"Ù7"ù 'ÉM'ù>'Â &" "l
20.0% 15.0%
12.5% 18.0%
- Q.172.** 'YK•, "%0"é6"ð 8"é"é0'2 ,"ù/"é '©0 11 "Y0"Ù7"1 'éG•" 8"Ù5'ù 'Y@ 3 'yA'%"@ "TMK 'É>'l@ "TMH-B 5"é0"Ù7"ù 'ÉM'ù>'Â &" "l
18.2% 15.0%
12.5% 18.0%
- Q.173.** 'YK•, "%0"é6"ð 8"é"é0'2 ,"ù/"é '©0 12 "Y0"Ù7"1 'éG•" 8"Ù5'ù 'Y@ 3 'yA'%"@ "TMK 'É>'l@ "TMH-B 5"é0"Ù7"ù 'ÉM'ù>'Â &" "l
16.7% 15.0%
12.5% 18.0%
- Q.174.** 'YK•, "%0"é6"ð 8"é"é0'2 ,"ù/"é '©0 13 "Y0"Ù7"1 'éG•" 8"Ù5'ù 'Y@ 3 'yA'%"@ "TMK 'É>'l@ "TMH-B 5"é0"Ù7"ù 'ÉM'ù>'Â &" "l
15.4% 15.0%
12.5% 18.0%

- Q.175.

‘YK•, "%0“é6“ò 8“é““é0’2 ,”Ù/“é ’©0 14 “Y0”Ù7”¹ ’éG•" 8”Ù5’ù ‘Y@ 3 ‘yA’%0@ “™K ‘É>’I@ “™H–B 5“é0”Ù7“ù ’ÉM’ù>‘Â &“

14.3%

15.0%

12.5%

18.0%
- Q.176.

‘YK•, "%0“é6“ò 8“é““é0’2 ,”Ù/“é ’©0 5 “Y0”Ù7”¹ ’éG•" 8”Ù5’ù ‘Y@ 3 ‘yA’%0@ “™K ‘É>’I@ “™H–B 5“é0”Ù7“ù ’ÉM’ù>‘Â &“ ”

40.0%

15.0%

12.5%

18.0%